

Merkle Science: Securing the blockchain and cryptocurrencies with Google Cloud

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ABOUT MERKLE SCIENCE

With Google Cloud, Merkle Science is helping businesses and organizations use cryptocurrencies, blockchain technology, and artificial intelligence to meet their compliance obligations and law enforcement and regulatory authorities to identify suspect



MERKLE SCIENCE

About Merkle Science

Merkle Science provides blockchain transaction monitoring and intelligence solutions for cryptoasset service providers, financial institutions, and government agencies to detect, investigate, and prevent money laundering, terrorist financing, and other criminal activities. Merkle Science is headquartered in Singapore with offices in Bangalore, Tokyo, and Seoul and is backed by Digital Currency Group, Kenetic, SGInnovate, and LuneX.

transactions. Google enabled the business about three months the time to develop products and services combine public data BigQuery with proprietary algorithms and infrastructure solutions to provide analysis and intelligence to customers.

Industries: Financial Services & Insurance

Location: Singapore

Storage (<https://cloud.google.com/storage/>),

Google Cloud (<https://cloud.google.com/>) flow/),

Cloud Bigtable

BigQuery (<https://cloud.google.com/bigquery/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Cloud Composer

(<https://cloud.google.com/composer/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Dataflow (<https://cloud.google.com/dataflow/>)

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Cloud Bigtable (<https://cloud.google.com/bigtable/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

App Engine (<https://cloud.google.com/appengine/>)

AI Platform (<https://cloud.google.com/ai-platform/>)

Google Cloud results

- Moves to near real-time streaming of events into data warehouse to provide a more timely service to customers
- Improves the efficiency of allocating

Decreased time to develop new products and services by up to one year

engineering

resources

- **Helps
businesses,
law
enforcement,
and
cryptocurrency
exchanges
thwart
financial crime**

Founded in 2018, **Merkle Science**

(<https://merklescience.com/>) provides infrastructure that helps businesses—including cryptocurrency exchanges, investment funds, banks and other companies—and government agencies using blockchain technology to thwart criminal activity and financial crime. Headquartered in Singapore, Merkle Science is backed by venture capitalists Digital Currency Group, Kenetic, LuneX Ventures and the Singapore Government-supported deep technology fund SGInnovate.

Merkle Science emerged from an incubator that connects skilled individuals who aim to build businesses. “I joined the group to leverage my background in fraud analytics and sciences, specializing in online payment fraud detection,” says Nirmal AK, Chief Technology Officer of Merkle Science. At the incubator, Nirmal met Mriganka Pattnaik, who had identified a sizable problem: the close focus of authorities and law enforcement on money laundering through crypto exchanges and

blockchain businesses, paired with the general lack of regulation.

“Law enforcement bodies were scrutinizing bank accounts and shutting down exchanges and businesses that did not have anti-money-laundering policies in place—meaning markets created over years were disappearing during shutdowns that could last six months or longer,” says Pattnaik. “So compliance was going to be a key issue for the sector going forward.”

Nirmal and Mriganka began conversations about collaborating and talked to a range of prospective customers before launching Merkle Science in November 2018. Between the startup date and May 2019, the founders focused on obtaining rounds of seed funding from venture capitalists and building the business. From three foundation employees in Singapore, Merkle Science has added about 22 employees in Bangalore, Tokyo, and Seoul.



The Merkle Science team at SGInnovate

Meeting compliance obligations

Merkle Science focuses on building products that help businesses using cryptocurrency and blockchain technology meet compliance obligations.

“We data mine the blockchain—the largest publicly available economics dataset in the world—and scrape large parts of the internet for information relevant to cryptocurrency policing,” says Nirmal. “So cryptocurrency exchanges can use our platform to send us customer transactions for risk screening. We then conduct extensive analysis to identify risks associated with transactions, users, or platforms.”

These risks may include whether a transaction is associated with payments on darknet markets (restricted-access online markets commonly used to buy and sell illicit goods), involves funds stolen from hacks of cryptocurrency exchanges, or is associated with a Ponzi scheme.

Merkle Science’s second product is a forensic tool that uses graphical network analysis to help law enforcement agencies deanonymize identities. “If an agency wants to investigate specific transactions, it can use the tool to track potentially illicit flows of funds,” says Nirmal. “So, for example, if alleged criminals are using darknet marketplaces to liquidate funds, we can locate the transactions and the exchanges through which these occurred.”

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—Nirmal AK, Chief
Technology Officer,
Merkle Science

Solid customer uptake

The business hit a sweet spot, its products experiencing solid uptake. “We are generating considerable revenue and have several potential clients coming forward,” says Nirmal. “The response from law enforcement in particular has been good—it’s always looking for tools like the ones we offer.”

Merkle Science’s data-oriented business model aligned closely with the products, services, and culture of Google Cloud (<https://cloud.google.com/>) too. Nirmal worked closely with Google Cloud to expand the BigQuery ([/bigquery](https://cloud.google.com/bigquery)) public datasets created for the Bitcoin blockchain and other blockchain-based currencies. “As someone who

contributed to publishing these datasets, using BigQuery to launch the business was a no-brainer,” says Nirmal.

The fledgling business received extensive support from Google Cloud account and technical teams, including inputs into the architecture that would best meet its needs and how best to scale.

Merkle Science is using [Cloud Composer](#) (/composer) for managed workflow orchestration and authoring, scheduling and monitoring of pipelines. “Tools like Cloud Composer allow us to ‘hit sixes from the very first ball’ as a business,” says Nirmal. “We love the product, and it manages one of the most important functions our team carries out.”

In addition, the business can complement the public cryptocurrency datasets on BigQuery with a proprietary dataset collected from multiple data sources. “We feed information from mining, crawling, and third parties into a data ingestion pipeline, with messages sent through [Pub/Sub](#) (/pubsub) and [Dataflow](#) (/dataflow) on to BigQuery,” says Nirmal. “We then combine these two datasets and come up with custom heuristics and algorithms to extract value, including a clustering algorithm that enables us and our customers to say, for example, ‘all these bitcoin user names probably belong to the same user.’”

Merkle Science also uses [App Engine](#) (/appengine) to conduct application deployments and [Cloud Storage](#) (/storage) to store files, while [Cloud Bigtable](#) (/bigtable) provides the throughput needed for the

organization's NoSQL database. "We rely heavily on Cloud Bigtable for its speed and scalability, and simply cannot replace its functionality," says Nirmal.

"The managed services available from Google Cloud meant we could focus on the business rather than on managing underlying infrastructure."

*—Nirmal AK, Chief
Technology Officer,
Merkle Science*

Focus resources on development

Google Cloud managed services enable Merkle Science to focus its engineers on customer service, uptake, and rapid application development rather than administration and maintenance tasks. "The managed services available from Google Cloud

meant we can focus on the business rather than on maintaining underlying infrastructure,” says Nirmal. “For one of our key applications, I could spend just one day on planning and early development and a second day on writing the core. I believe the managed services decreased the time needed to develop our products and services by about three months.”

BigQuery and Cloud Identity and Access Management (/iam) also enable the business’s scientists and analysts to gain secure, role-based access to data for analysis.

“Google Cloud is
key to our
growth and we
envisage its
products and
services
continuing to
play an
important role in
our
development.”

—Nirmal AK, Chief
Technology Officer,
Merkle Science

Further optimization underway

The business is moving to further optimize its Google Cloud deployment by moving its applications to [Google Kubernetes Engine](#) (/kubernetes-engine) to exploit features such as fast build-and-deploy, monitoring, and support from expert Google Site Reliability Engineers and enhanced workload and network security. Nirmal and his team are experimenting with Google Cloud artificial intelligence (AI) tools and expect to scale up their usage in the near future. “There is a Data Labeling Service built into our AI Platform that is potentially extremely useful, as is the AI Platform Pipelines product currently in beta,” says Nirmal. “These services—plus the fact we have everything in one place—can help save our business costs and efforts.”

Merkle Science is now ideally positioned to grow as the blockchain and digital currencies assume greater importance in markets around the world. “Google Cloud is key to our growth and we envisage its products and services continuing to play an important role in our development,” says Nirmal.